

- 6 (a) (i) at 22.5°C , $R_T = 1600\Omega$ or $1.6\text{ k}\Omega$ C1
total resistance = 800Ω A1 [2]
- (ii) *either* use of potential divider formula *or* current = $9 / 2000$ (4.5 mA) C1
 $V = (0.8/2.0) \times 9$ $V = (9/2000) \times 800$
 $= 3.6\text{ V}$ $= 3.6\text{ V}$ A1 [2]
- (b) (i) total resistance = $4/5 \times 1200$ C1
 $= 960\Omega$ A1 [2]
- (ii) for parallel combination, $1/960 = 1/1600 + 1/R_T$ C1
 $R_T = 2400\Omega$ / $2.4\text{ k}\Omega$
temperature = 11°C A1 [2]

Page 4	Mark Scheme: Teachers' version	Syllabus	Paper
	GCE A LEVEL – October/November 2010	9702	22

(c) e.g. only small part of scale used / small sensitivity
non-linear

B1

B1

[2]

(any two sensible suggestions, 1 each, max 2)

7 (a) (i) most α -particles were deviated through small angles

B2

[2]